

Automated Hyper-Parameter Optimization for Eye Movement Artifact Removal

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Abstract—Brain-computer interface (BCI) systems allow users to control external devices with their brain waves. However, electroencephalography (EEG) signals used by most BCI systems are prone to artifacts from various sources (e.g., muscle activity, eye movements, and electrical interference). These artifacts can degrade the performance and usability of BCI systems. Many tools exist to eliminate these artifacts. However, not all methods are automated, and some might require tuning certain hyper-parameters for optimal performance. We propose a method to automatically optimize the hyper-parameters of an eye blink artifact removal tool to improve the removal of artifacts in resting state EEG. We use a subset of eye movement artifacts to optimize the hyper-parameters using the EEG Quality Index (EQI) as the objective function. The optimized hyper-parameters are then used in a test artifact to quantify the improvement of the EQI. Results show improvement in the EQI when compared to the default artifact removal hyper-parameters, and raw EEG traces. We conclude that our method can provide a personalized and robust artifact removal solution for BCI users with complex needs.

Index Terms—EEG, artifact removal, automated, eye movement

I. INTRODUCTION

BCI systems allow an individual to control external devices (e.g., a computer, a wheelchair, a communication device) using their brain waves [1]. Most common BCI systems use non-invasive EEG to measure the electrical potential of the brain using sensors that are placed on the surface of the scalp. Due to the distance between where the signal originates (at the neuronal level) and sensor placement, there is significant opportunity for noise to be introduced to the EEG signal as it propagates through many physiological layers such as brain tissue, cerebral spinal fluid, skull, and scalp [2]. Additionally, the skull introduces electrical isolating properties, effectively attenuating and low-pass filtering the EEG signal significantly, resulting in only very small electric potentials picked up by the scalp sensors (i.e., on the order of microvolts) [3]. The EEG frequency range also falls within the range of many potential noise sources (e.g., electrical interference, muscle activity, blinking, and heart rate). Thus, effective artifact removal

algorithms are of great interest to the BCI community, with substantial potential to impact outcomes for BCI use [4].

One significant challenge in artifact removal processing of EEG data is that most artifacts depend on non-linear and quasi-stochastic sources (e.g., eye movements, muscle artifacts) [5]. These noise sources vary between subjects as they can be caused by different physiological sources such as movements (e.g., limbs, or eye movement), other bioelectrical potentials (e.g., heart beat, blinking) and fluctuations in skin resistance [6]. As individuals with complex needs represent a population that could significantly benefit from advancements in BCI technology, better methods for removing these types of artifacts can help facilitate the use of BCI by such end-users.

Current automated artifact removal methods use hyper-parameters which can drastically alter their effectiveness [5]. These hyper-parameters tend to have suggested ranges included in their implementation, but little has been emphasized on how those suggested ranges were developed. Usually, the hyper-parameters are not fine-tuned for each individual. Instead, there is a prevailing tendency to rely on values reported in previous literature, even though there may be considerable differences across studies in setup, equipment, experimental conditions, or other environmental factors. The purpose of this project is to test the viability of an automated hyper-parameter optimization methodological processing pipeline to set the hyper-parameters of an eye movement artifact removal tool to improve the quality of the measured EEG signal. This work could provide researchers and users a simple method to remove artifacts using a personalized approach, thereby improving the BCI experience for individuals whose disability might cause involuntary movements that affect the EEG quality an desired system performance.

II. METHODOLOGY

A. Artifact removal tool

Methodology for cleaning single channels of EEG data using k-means and Singular Spectrum Analysis (SSA) as described by Maddirala and Veluvolu [7] was first extended to work on multiple channels, as an artifact removal tool (ART). The workflow of the tool is as follows (Fig. 1). The original method takes a single EEG trace, and constructs an embedded matrix of a set window length. Kurtosis, Hjorth mobility, and

This work was funded by the HBI Brain CREATE and ACHRI Haslam postdoctoral fellowships.

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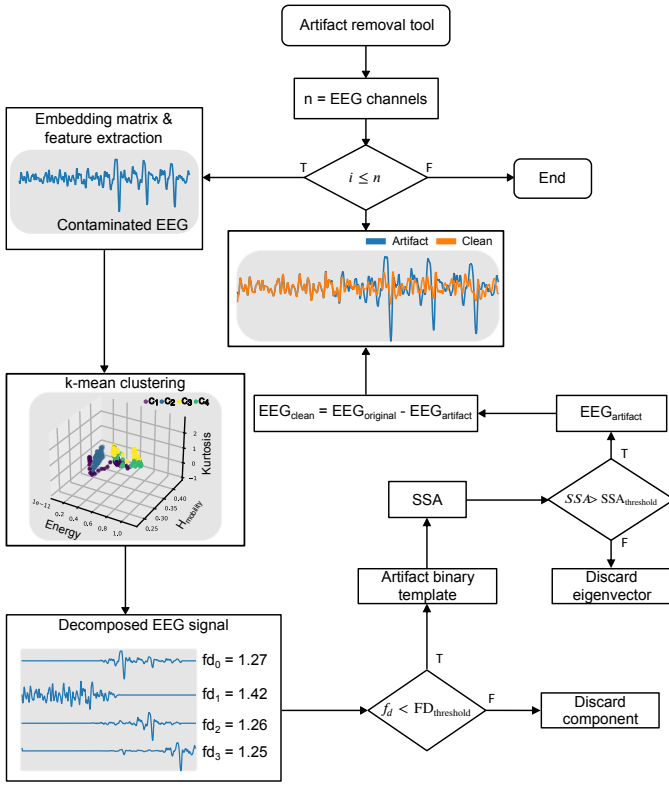


Fig. 1. Artifact removal tool pipeline, adapted from Maddirala and Veluvolu [7] to work with multi-channel EEG data.

energy features are computed for each window. A k-means classifier is used to create k-components from the previously mentioned features. Fractal dimension (FD) is computed for each reconstructed component and only the components below a set threshold (i.e., $FD_{threshold}$) are kept. These components are associated with high-amplitude, slow-varying (i.e., low frequency) signals. Reconstruction of the signal with these components provides an eye-blink or eye-movement binary artifact template [7]. The original signal (i.e., raw EEG) is then indexed using this binary template to keep only data samples that are associated with an artifact. However, the masked template signals still may include some EEG activity. The resulting signal is converted into an embedded matrix again, so the SSA can be calculated alongside a singular value decomposition (SVD) of the embedded matrix. The indices of the singular values whose ratio were greater than a preset threshold (i.e., $SSA_{threshold}$) are identified. The signal is then reconstructed to provide a template of the artifact without any EEG information. Lastly, the artifact template is subtracted from the raw EEG to obtain a clean, artifact-free trace. The extension of this methodology for the ART was to then repeat this process per channel, as needed.

B. EEG quality index

The EEG quality index (EQI) used in this work was built based on the outline presented by Fickling et al. [8]. The original implementation was extended to use any desired

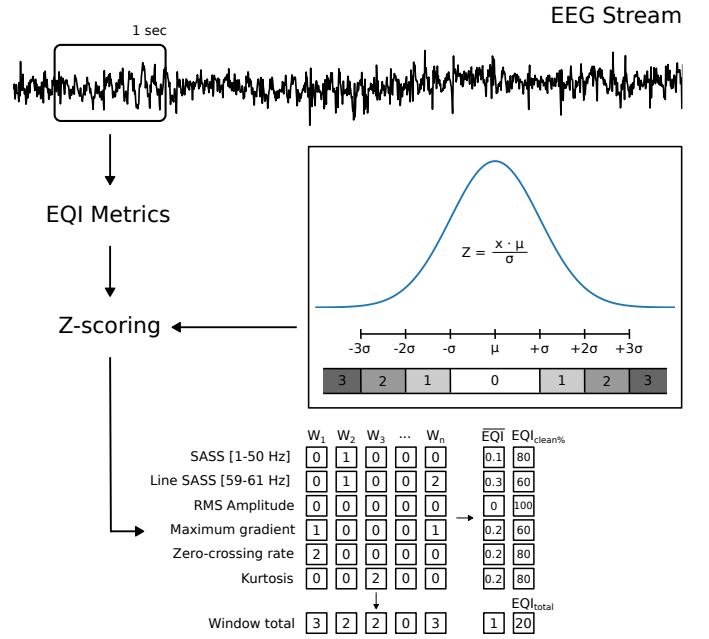


Fig. 2. EEG quality index z-scoring system obtained from a sliding window. The output variables are: EQI total, EQI clean percent, and EQI mean. Adapted from [8].

window size, sliding windows, and to use reference and test EEG traces with different sampling rates. The EQI consists of six different metrics to assess the similarity between a test and a reference (i.e., clean) EEG dataset. The metrics are calculated for a sliding window and a z-scoring system is used to weight the deviations of the test from the reference EEG. The metrics used for the EQI assessment are shown in Table I. The output per channel of the EQI includes an EQI total percentage ranging from 0 to 100 %, where 100 % signifies that the test reference falls within ± 1 z-score of the reference z-score (i.e., higher numbers indicate greater similarity to the reference EEG); an EQI clean percent where the percentage of clean windows (i.e., within ± 1 z-score) per metric (higher values indicating more clean windows in the sample evaluated); and an EQI mean with the averages across windows for each metric (Fig. 2).

C. Optimization

The extended ART and EQI were implemented in Python 3.11 using the Numpy (v1.26) and Scipy (v1.12) packages [9]. The dataset used was the Temple University Hospital EEG Artifact Corpus (v2.0) [10] which contains labeled EEG traces of various artifacts (i.e., eye movement, chewing, shivering, electrode pop, electrode static and lead artifacts, and muscle artifacts). For this project, we focused on eye movement artifacts due to their prevalence and relevance for BCI applications. Custom scripts were used to trim the EEG traces according to the Temple dataset annotations. Unlabeled segments of the data were considered to be clean, non-artifact background activity.

TABLE I

METRICS CONSIDERED FOR THE EEG QUALITY INDEX COMPUTATION. EACH METRIC IS COMPUTED FOR A TEST AND A REFERENCE (I.E., CLEAN) EEG TRIAL. EACH METRIC IS COMPUTED FOR MOVING WINDOWS OF THE INDIVIDUAL EEG CHANNELS. ADAPTED FROM [8].

EEG Quality Index metric	Equation
Average single sided amplitude spectrum (SSAS)[1-50 Hz]	$Y(k) = \sum_{n=1}^N X(n)W_n^{(n-1)(k-1)}$
Line noise SSAS [59-61 Hz]	Sames as SSAS but between 59 and 61 Hz
RMS Amplitude	$x_{RMS} = \sqrt{\frac{1}{N} \sum_{n=1}^N x_n ^2}$
Maximum gradient	$MG = \max[x(n) - (x(n-1))]$
Zero-crossing rate	$ZCR = \frac{1}{N} \sum_{n=1}^N \text{signum}[x(n)] - \text{signum}[x(n-1)] $
Kurtosis	$\text{Kurtosis} = \frac{\frac{1}{N} \sum_{i=1}^N (x(i) - \bar{x})^4}{(\frac{1}{N} \sum_{i=1}^N (x(i) - \bar{x})^2)^2}$

The optimization workflow is as follows. Eye movement artifacts, and clean segments with a minimum duration of 3 sec were extracted for a single trial. A total of 8 trials were used. From each trial, the artifacts were divided into optimization (n=5) and testing (n=1) categories. The optimization subset was used to optimize the hyper-parameters of the ART (i.e., number of k-means clusters ($n_{clusters}$), $FD_{threshold}$, and $SSA_{threshold}$) using a standard simulated annealing meta-heuristic search method with initial annealing temperature of 1.0, linear multiplicative cooling schedule with fixed rate of 0.99, and Metropolis acceptance criterion [11]. The objective function was defined as the reciprocal of the average EQI across all artifact channels associated with the artifact in question (i.e., $100 - EQI_{total \text{ percentage}}$). The reciprocal was used to pose the objective function as a minimization problem as EQI provides a metric from 0 to 100 % where 100 % represents a clean signal (i.e., a signal where the EQI features lie within 1 z-score from the reference signal), thus requiring the reciprocal to optimize for the cleanest signal possible. The values of the hyper-parameters that yielded the lowest objective function were saved as that trial's optimal hyper-parameters. Next, the optimal hyper-parameters were used to compute the EQI of the test artifacts using the same clean segments used in the optimization step. As a reference point, the EQI of the withheld test subset was computed using "default" hyper-parameters described by Maddirala and Velovolu [7] (i.e., $n_{clusters} = 4$, $FD_{threshold} = 1.4$, and $SSA_{threshold} = 0.001$). Additionally, the EQI was also computed using the test artifacts without any additional artifact removal techniques (i.e., raw signals).

III. RESULTS

The distribution of the values of the optimized hyper-parameters shows a wide range of values (Fig. 3). The value that deviated the most from those suggested by Maddirala and Velovolu is the fractal dimension threshold, where the optimization step found improved outcomes using an $FD_{threshold}$ with a mean of 6.48, as compared to the default value of 1.4 [7].

The EQI between the optimized hyper-parameters shows overall higher values (i.e., test EEG traces are more similar to

the reference EEG traces) than the default parameters or the raw signals (Fig. 4).

IV. DISCUSSION

The optimized values for the $FD_{threshold}$ shows higher values than the ones suggested by Maddirala and Velovolu (i.e., 1.4) [7]. Since the FD value is related to the length of the signal, it implies that the artifacts in the selected dataset are either of higher amplitude, or higher frequency content than the eye blink artifacts of the original paper. Similarly, the $SSA_{threshold}$ values found through the optimization step are greater than the suggested 0.01 default value for every fold tested. Even the lowest optimized $SSA_{threshold}$ value was double the suggested 0.01. The $SSA_{threshold}$ is the ratio of singular values normalized by the sum of total singular values. Only those singular values whose ratio is greater than the threshold are utilized in reconstructing the artifact signal. Thus, a higher ratio value would require the singular values of interest to be greater, making it more "conservative" to what is considered signal of interest (i.e., higher $SSA_{threshold}$ equals more signal considered to be noise).

The optimization approach suggested here improves the quality of the signal as shown in the increase of the EQI computed using the optimized hyper-parameters vs the default values and the raw artifact signals (Fig. 4). However, not all channels and EQI metrics showed improvement (i.e., positive values). Suggesting that even with the optimization of the hyper-parameters, the artifact removal method is quite dependent on artifact topology. One limitation of the current optimization method employed is that we are only considering the hyper-parameters that produced the overall lowest objective function (i.e., using a single artifact), but hyper-parameters will vary for each artifact of interest used in the optimization. Thus, future work could include exploring a scheme in which the optimized hyper-parameters of all artifacts are considered jointly for a more comprehensive optimization using some weighted average implementation or similar.

Another limitation of the proposed approach is the need for labeled segments of clean and artifact data. In the validation shown here, we used the Temple University Artifact Corpus

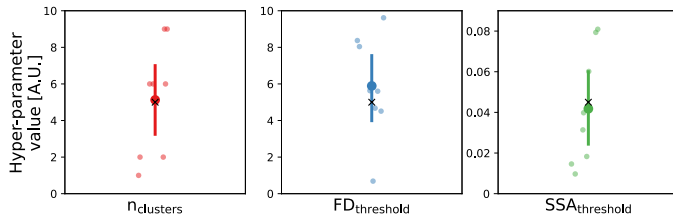


Fig. 3. Distribution of the values of the hyper-parameters after optimization. Each hyper-parameter shows the median (large circle) $\pm$ standard deviation (vertical lines). Default values are shown as with the black "x" marker. The smaller circles are the values of the optimized hyper-parameter of each trial (n=8).

EEG dataset as this dataset already includes labels for clean data segments and various types of artifacts. However, as discussed in the introduction, the interest of an automated artifact removal tool is in its potential for implementation within a BCI system. The need for these labeled segments of clean and artifact data would make the implementation unfeasible if an EEG expert is not available during data acquisition for identifying clean versus artifact data, thus limiting current use to in-lab experiments. Future implementations could set predefined parameters to automatically label the clean and artifact segments; this type of implementation should be validated against the current state-of-the-art approach where an expert manually labels the segments.

Lastly, the work presented here was all performed offline with a resting state dataset. Future validations should be implemented in a real-time BCI experiment where some baseline recording is taken to optimize the hyper-parameters. Then, the optimized hyper-parameters can be used to clean the raw EEG traces to see if the BCI performance (e.g., classification accuracy) is improved. Additionally, the time that it takes to do the optimization protocol should be considered, as long times can negatively impact the experience for the user. Currently, the hyper-parameter optimization process takes about 2 minutes to run on 5 different artifacts when running on a moderately computationally capable computer (AMD Ryzen 7 5800HS with 16 GB of DDR4 RAM). This time can also quickly increase with more channels, making it less desirable for high-density EEG headsets.

V. CONCLUSION

We have presented a new methodology to optimize hyper-parameters of an artifact removal tool by using the EEG quality index metric as the objective function. This approach could make it possible for individuals whose disability causes frequent artifacts (e.g., eye blinks, eye movements, muscle activity) to improve their signal quality when using BCI systems. For initial validation of the methodology, we have used the well-labelled Temple University Artifact Corpus EEG dataset. We have demonstrated that the EEG quality index improves for the optimized hyper-parameters in comparison to the recommended default parameters for artifact removal, as well as no artifact removal (i.e. raw artifact data). Future validations should aim to use this methodology with real-time

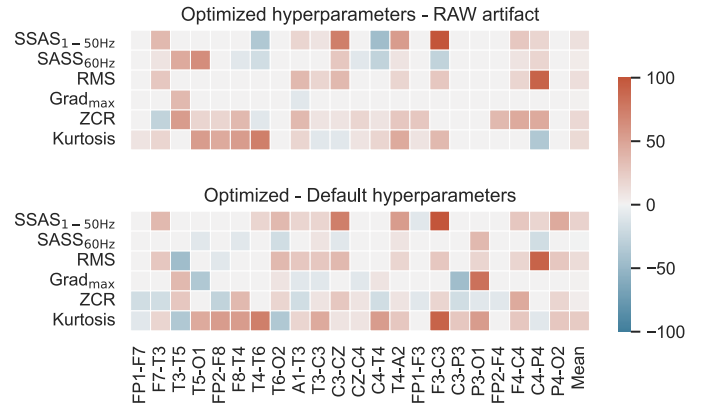


Fig. 4. Representative differential EEG Quality Index of a single trial computed for the optimized hyper-parameters vs raw signal (top), and the optimized hyper-parameters vs default hyper-parameters (bottom). Positive values (i.e., red) signify that the optimized hyper-parameters achieved a cleaner EEG signal than the raw or default hyper-parameters. Negative values (i.e., blue) signify that the raw or default hyper-parameters achieved a cleaner EEG signal than the optimized hyper-parameters.

experiments. Additionally, other artifact removal algorithms could implement a similar optimization approach to improve artifact rejection on a per-subject basis.

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